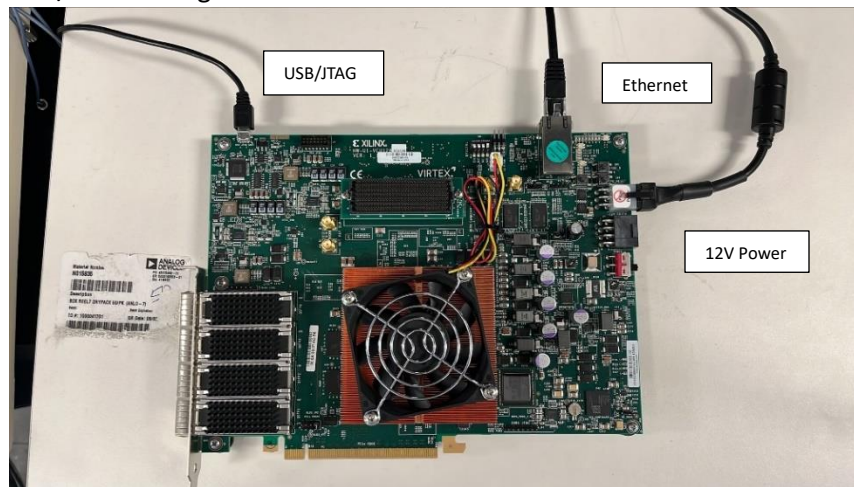


1. Software Setup

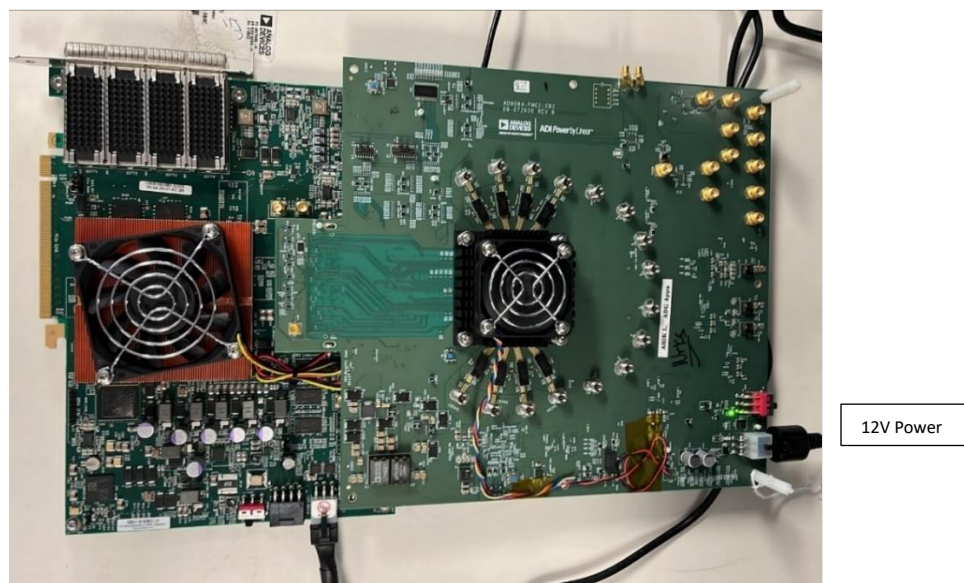
- a. Install Xilinx Software Version 2021.2 or later
<https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/vivado-design-tools/archive.html>
- b. Install cable drivers <https://docs.amd.com/r/en-US/ug973-vivado-release-notes-install-license/Installing-Cable-Drivers>
- c. If not using precompiled binaries setup cross compile environment using the Microblaze toolchain: microblazeel-xilinx-linux-gnu-gcc

2. Hardware Setup

- a. VCU128:
 - i. Connect 12V Power
 - ii. Connect Ethernet to host PC
 - iii. UBS/JTAG through micro usb to host PC



- iv.
- b. Connect Apollo CE (FMCC or FMCA) internally clocked board to VCU128
 - i. FMC Riser necessary
 - ii. Connect 12V Power



- iii.
- c. Power on both VCU128 and Apollo Board

3. VCU128 Startup

- a. Open terminal to start hardware server:

```
PS C:\Users\rbear> <path_to_Xilinx>\Xilinx\HWSRVR\2021.2\bin\hw_server
```

```
***** Xilinx hw_server v2021.2
**** Build date : Oct 19 2021 at 03:13:30
** Copyright 1986-2021 Xilinx, Inc. All Rights Reserved.
```

```
INFO: hw_server application started
INFO: Use Ctrl-C to exit hw_server application
```

INFO: To connect to this hw_server instance use url: **TCP:<host_name>:3121**

- b. Open second terminal to load fpga and kernel:

```
PS C:\Users\rbear> <path_to_Xilinx>\Xilinx\Vitis\2021.2\bin\xsct
```

```
***** Xilinx Software Commandline Tool (XSCT) v2021.2.0
**** SW Build 1967 on 2021-10-14-04:42:58
** Copyright 1986-2021 Xilinx, Inc. All Rights Reserved.
```

```
xsct% connect -host <host_name> -port 3121
```

```
xsct% targets
```

```
1 xcvu37p
2 MicroBlaze Debug Module at USER2
3* MicroBlaze #0 (Running)
4 Legacy Debug Hub
```

```
xsct% fpga -f <path_to_image>/ad9084_vcu128.bit
```

```
100% 81MB 1.7MB/s 00:48
```

```
xsct% targets
```

```
1 xcvu37p
2 MicroBlaze Debug Module at USER2
5 MicroBlaze #1 (Running)
3* MicroBlaze #0 (Running)
4 Legacy Debug Hub
```

```
xsct% rst -processor -clear-registers -stop
```

```
Invalid clear registers request
```

```
Info: MicroBlaze #0 (target 3) Stopped at 0x0 (External debug request)
```

```
xsct% target 5
```

```
xsct% rst -processor -clear-registers -stop
```

```
Invalid clear registers request
```

```
Info: MicroBlaze #1 (target 5) Stopped at 0x0 (External debug request)
```

```
xsct% dow <path_to_linux_image>/simpleImage.vcu128_ad9084_adg.strip
```

```
Downloading Program --
```

```
<path_to_linux_image>/simpleImage.vcu128_ad9084_adg.strip
```

```
section, .text: 0x80000000 - 0x80686407
```

```
section, __fdt_blob: 0x80686408 - 0x80696407
```

```

section, .rodata: 0x80697000 - 0x80c02d27
section, .pci_fixup: 0x80c02d28 - 0x80c04c87
section, .builtin_fw: 0x80c04c88 - 0x80c04cab
section, __ksymtab: 0x80c04cac - 0x80c101a7
section, __ksymtab_gpl: 0x80c101a8 - 0x80c1a53f
section, __ksymtab_strings: 0x80c1a540 - 0x80c3bb2a
section, __param: 0x80c3bb2c - 0x80c3c233
section, __modver: 0x80c3c234 - 0x80c3c2c3
section, __ex_table: 0x80c3c2d0 - 0x80c3d7ef
section, .notes: 0x80c3d7f0 - 0x80c3d843
section, .sdata2: 0x80c3e000 - 0x80c3dfff
section, .data: 0x80c3e000 - 0x80cd25bf
section, .data..percpu: 0x80cd3000 - 0x80cd2fff
section, .init.text: 0x80cd3000 - 0x80d02feb
section, .init.data: 0x80d02fec - 0x80d06367
section, .init.ivt: 0x80d06368 - 0x80d0638f
section, .init.setup: 0x80d06390 - 0x80d06827
section, .initcall.init: 0x80d06828 - 0x80d06d87
section, .con_initcall.init: 0x80d06d88 - 0x80d06d8b
section, .bss: 0x80d07000 - 0x80d2cf97
section, .init.ramfs: 0x80e2d000 - 0x81479e23
100% 19MB 0.2MB/s 01:44
Setting PC to Program Start Address 0x80000000
Successfully downloaded
<path_to_linux_image>/simpleImage.vcu128_ad9084_adg.strip
xsct% targets
1 xcvu37p
2 MicroBlaze Debug Module at USER2
5* MicroBlaze #1 (External debug request)
3 MicroBlaze #0 (External debug request)
4 Legacy Debug Hub
xsct% con
Info: MicroBlaze #1 (target 5) Running
xsct% target 3
xsct% con
Info: MicroBlaze #0 (target 3) Running
xsct%

```

- c. Connect using PuTTY
 - i. Create serial connection to proper COM port; set Baud to 115200
 - ii. Login: **root** ; Password: **analog**
 - iii. Set IP to desired address: **ifconfig eth0 192.168.0.11 netmask 255.255.255.0;**
ifconfig eth0 down; ifconfig eth0 up
 1. Verify adapter settings match
- d. Connect over ssh
 - i. **ssh root@<ip>**

4. Running Examples

- a. Examples that have been validated:
 - i. ads10_apollo_ex_main/
 - 1. fullchip
 - 2. fullchip_8t8r
 - 3. fullchip_hop (partial functionality see known issues)
 - 4. bysnc_tof
 - 5. mcs_cal
 - 6. prbs
 - 7. rx_jesd
 - 8. rx_adc_bmem
 - 9. fullchip_mcs_sc1_dl (sc1 profile required)
 - a. use SMA J12 on the VCU128 to view pattern start output trigger
- b. Cross compiling

- i. Open git bash or similar program (run as admin might be necessary)
 - ii. Check you have the Microblaze toolchain in your path:
"microblazeel-xilinx-linux-gnu-gcc -v"

```
# cd <path>/apollo_REL_<ver>/pkg/src/examples
# cd ads10_apollo_ex_<example>
# make clean all PLATFORM=VCU128 CROSS_COMPILE=microblazeel-xilinx-linux-gnu-
```

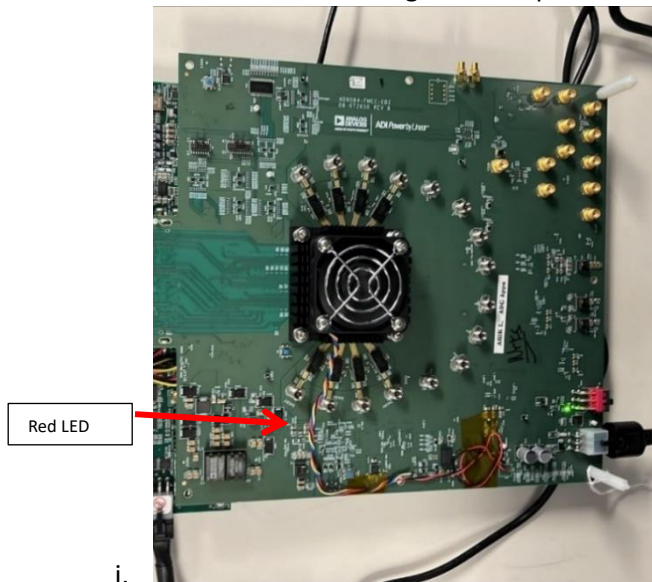
- iii. Copy executable + fw to VCU128 with the following file structure:

```
root
|-apollo_api_dev2
| |-examples
| | |-fw_images
| | | |-b0
| | | | |-app_signed_encrypted_B
| | | | |-flash_image_0x21000000.bin
| | | | |-flash_image_0x20000000.bin
| | | | |-flash_image_0x02000000.bin
| | | | |-flash_image_0x01030000.bin
| | | | |-APOLLO_FW_CPU1_B.bin
| | | | |-APOLLO_FW_CPU0_B.bin
| | |-ads10_apollo_ex_<example>
| | | |-debug
| | | | |-apollo_main
```

5. Known Issues

- a. There is a power fault that can occur on internally clocked Apollo FMCC boards during the first run or if the example is ran too soon after boot. It can be observed by either a red LED lighting up on the Apollo CE Board or the following error when running an example:

Error in `adi_ads10_apollo_ex_startup`. To resolve, simply reboot the Apollo CE Board and wait 15 seconds before running the example.



- i.
- b. If PRBS example is ran, VCU128 needs to be re programmed to successfully run `fullchip_8t8r` or else links will not come up.
- c. Examples not listed in this document are not guaranteed to be functional.
- d. Tmaster has a bug that makes it hang when using hsci and triggers at the incorrect time using spi.